Semi-Supervised Semantic Segmentation for Remote Sensing Images via Multiscale Uncertainty Consistency and Cross-Teacher–Student Attention

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***Abstract*— Semi-supervised learning offers an appealing solution for remote sensing (RS) image segmentation to relieve the burden of labor-intensive pixel-level labeling. However, RS images pose unique challenges, including rich multiscale features and high interclass similarity. To address these problems, this article proposes a novel semi-supervised multiscale uncertainty and cross-teacher-student attention (MUCA) model for RS image semantic segmentation tasks. Specifically, MUCA constrains the consistency among feature maps at different layers of the network by introducing a multiscale uncertainty consistency (MSUC) regularization. It improves the multiscale learning capability of semi-supervised algorithms on unlabeled data. In addition, MUCA utilizes a cross-teacher–student attention (CTSA) mechanism to guide the student network, guiding the student network to construct more discriminative feature representations through complementary features from the teacher network. This design effectively integrates weak augmentation (WA) and strong augmentation (SA) to further boost segmentation performance. To verify the effectiveness of our model, we conduct extensive experiments on ISPRS-Potsdam and LoveDA datasets. The experimental results show the superiority of our method over state-of-the-art semi-supervised methods. Notably, our model excels in distinguishing highly similar objects, showcasing its potential for advancing semi-supervised RS image segmentation tasks.**

***Index Terms*— Consistency regularization, deep learning, remote sensing (RS) images, semi-supervised semantic segmentation.**

I. Introduction

DEEP learning techniques on sensing images semantic segmentation [1], [2] provide promising solutions for disaster prevention [3], land-use surveillance [4], environment protection [5], and urban planning [6]. The increasing number of Earth-observation satellites makes a large amount of raw remote sensing (RS) images be continuously captured, which amplifies the advantage of deep learning methods [7]. However, it is a time-consuming and laborious task to manually label a large amount of RS data into a large number of categories with the complexity of labeling rules [8]. Therefore, semi-supervised learning draws the attention of the RS communities because it only uses a small number of labeled samples and a large amount of unlabeled [9].

Several semi-supervised semantic segmentation methods have been conducted for natural images, which are mainly divided into teacher-student consistency [10], feature perturbation consistency [11], and self-training pseudo-labeling [12]. Self-training pseudo-labeling methods often experience performance degradation due to incorrect pseudo-labels generated in the early stages, which can lead to error propagation. Feature perturbation consistency approaches rely heavily on selecting appropriate perturbations and balanced labeled data, making them particularly sensitive to data imbalance. Among these, the teacher-student consistency framework stands out as the most stable and is especially well-suited for tasks that demand high model complexity [13]. Recent works [9], [14] have introduced semi-supervised segmentation into the field of RS image analysis, meanwhile, encountered some domain-specific problems [15], e.g., rich multiscale information and high interclass similarities. Such inherent domain gaps between natural and RS images pose great challenges for semi-supervised RS image segmentation.

1) *Rich Multiscale Information [16]:* Objects on the RS image exhibit a wide range of scales, from expansive objects (e.g., Building) to small entities (e.g., Car). As shown in Fig. 1(a) and (b), the yellow regions represent cars while the blue regions denote buildings, demonstrating the significant size variations across different categories in RS images. Furthermore, objects within the same category exhibit substantial size disparities, as exemplified by the buildings illustrated in Fig. 1(c) and (d).
2) *High Interclass Similarities [17]:* Objects from different categories appear to have high visual similarity and are usually intertwined on the RS images. For example,

Received 13 March 2025; revised 24 April 2025 and 11 June 2025; accepted 30 June 2025. Date of publication 2 July 2025; date of current version 11 July 2025. This work was supported in part by the Science and Technology Development Fund, Macau, SAR, under Grant 0006/2024/RIA1; and in part by the National Natural Science Foundation of China under Project 61971388 and Project 62376237. *(Corresponding authors: Xin Sun; Changrui Chen.)*

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Digital Object Identifier 10.1109/TGRS.2025.3585489

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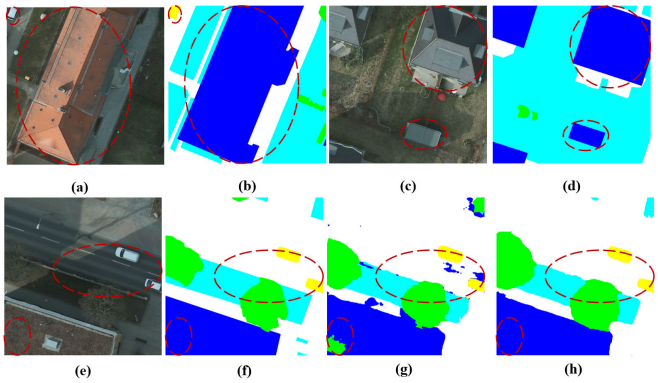


Fig. 1. (a)–(d) Image from the ISPRS-Potsdam dataset and its ground truth. (e)–(h) Image from ISPRS-Potsdam dataset, Ground Truth, and results from Unimatch and Ours.

as shown in Fig. 1(g), the boundary regions of Low vegetation, Tree, Car, and Building are misclassified with Unimatch [18].

To address the above challenges, we propose a new semi-supervised semantic segmentation model for RS images based on multiscale uncertainty and cross-teacher-student attention (MUCA).

1) For the first time, we propose a multiscale uncertainty consistency (MSUC) module. The module constrains the consistency among feature maps at different layers of the network model, which improves the multiscale learning capability of semi-supervised algorithms on unlabeled data. The traditional semi-supervised algorithms perform the consistent regularization between teacher and student only at the last layer, which ignores the rich information learned in the intermediate layers from the large amount of unlabeled data. In contrast, our proposed semi-supervised method can learn efficient and rich multiscale information from unlabeled RS images with our MSUC module.
2) To solve the problem of high interclass similarities, the key is to improve the feature extraction ability of the model. Therefore, we propose a cross-teacher-student attention (CTSA) module, which makes the teacher network to help the student reconstruct the outputs of the encoder. Specifically, it takes the encoder result of the student as the query and the result of the teacher as the key and value. The benefit of such cross-network attention is that deep features can be learned from both the student and teacher models. It can help segment the hard-to-distinguish categories in RS images. For instance, as shown in Fig. 1, our semi-supervised model [see Fig. 1(h)] can achieve more accurate segmentation results than the SOTA Unimatch method [see Fig. 1(g)] on RS images.

Notably, previous studies have partially addressed the scientific challenges of rich multiscale information and high interclass similarities in RS fully supervised semantic segmentation models [16], [19], [20], [21]. However, in the field of semi-supervised RS, our article is the first attempt to simultaneously address both the scientific challenges and provide an open-source implementation. We first propose the MUCA model from the viewpoint of computing MSUC regularization and using CTSA. It prioritizes architectural innovations for RS semi-supervised semantic segmentation, advancing semantic segmentation performance under limited data. The proposed model is nonintrusive and can be easily integrated into existing semantic segmentation models without changing the network structure of the model itself. Overall, the major contributions of this work can be summarized as follows.

1) We propose a novel semi-supervised model, MUCA, specifically to address the challenge of rich multiscale information and high interclass similarities in RS semantic segmentation.
2) To learn rich multiscale information, MUCA introduces MSUC regularization that constrains the consistency among feature maps at different layers of the network.
3) To distinguish the high interclass similarities, MUCA utilizes a new CTSA to guide the student network in reconstructing discriminative encodings by the teacher.
4) Extensive experimental results demonstrate the efficacy of our MUCA model on the LoveDA and ISPRS-Potsdam datasets, compared with some SOTA methods. The results show promising achievement in RS semi-supervised segmentation tasks and highlight the significance of our contributions.

The rest of this article is organized as follows. Section II briefly introduces the related work. In Section III, our model MUCA is proposed and discussed. Section IV shows the experimental results and compares them with other SOTA methods. Section V concludes this article.

II. Related Work

In recent years, the developments of deep neural networks have encouraged the emergence of a series of works on semi-supervised semantic segmentation of RS images. This section gives a short description of recent developments in related fields.

A. RS Semantic Segmentation

Semantic segmentation is to segment an image into regional blocks with certain semantic meanings [22], [23] and recognize the specific semantic category of each regional block. It commonly labels the image pixel by pixel [24], [25], [26], [27], [28]. Early semantic segmentation models, including FCN [29], SegNet [30], and U-Net [31], achieved remarkable performance in traditional semantic segmentation tasks. With the success of Transformers in vision in recent years, more and more semantic segmentation models are adopting the Transformers architecture [32]. Complex backgrounds as well as resolution variations are the main challenges in the semantic segmentation of RS images. In RS images, large intraclass and small interclass variations between objects make image feature characterization difficult. Workman et al. [33] argued that samples with high-resolution labels can be used to guide the training process in supervised learning using low-resolution labels. Region aggregation methods have also been used to improve the resolution of images. Quan et al. [34] utilized

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multiscale edge features obtained by differential difference of Gaussian (DoG) methods to improve the results of edge extraction. Li et al. [35] proposed a progressive recurrent neural network to remove RS images destriping. Zhong et al. [36] proposed a Transformer-based noise identification network model to help mitigate the over-segmentation phenomenon. However, semantic segmentation algorithms rely heavily on a large amount of training data, and it is difficult for these frameworks to achieve good results when labeled data is limited.

B. Semi-Supervised Semantic Segmentation

Semi-supervised semantic segmentation algorithms focus on how to better utilize large amounts of unlabeled data than supervised algorithms. Most semi-supervised semantic segmentation models use a basic convolutional neural network as the backbone and implement semi-supervised algorithms in three different strategies.

The first strategy is pseudo-labeling for self-training, which involves generating pseudo-labels for unlabeled images based on a previously trained model on labeled data. It trains the model with newly generated images with pseudo-labels [37], [38], [39]. The second strategy is to optimize the model by optimizing the consistent regularized loss function of feature perturbations [32], [40], [41], [42]. Such methods utilize data-enhancement techniques to apply perturbations directly to the input image. They force the model to predict the same labels for both original and enhanced images. Some feature-based perturbation methods add internal perturbations to the segmentation network, resulting in a modified feature [43]. UniMatch [18] took into account the nature of the semantic segmentation task and merged appropriate data augmentation into FixMatch [11], which has evolved into a concise baseline of semi-supervised semantic segmentation algorithms. The last strategy is teacher-student consistency [44]. The most classical method based on consistency regularization is the Mean teacher [10]. It considers the consistency between the predictions of the student and the teacher network. The weights of the teacher network are calculated from the exponential moving average (EMA) of the weights of the student's network. Recently, a few works have applied the teacher-student consistency method to the field of semi-supervision of RS images [9].

C. Semi-Supervised Semantic Segmentation in RS Images

Semi-supervised RS semantic segmentation has drawn more and more attention to make better use of the large amount of unlabeled RS image data [45], [46], [47], [48]. These RS semi-supervised semantic segmentation models can be broadly categorized into two groups.

The first one is to address the limitations of semi-supervised models. For instance, Lu et al. [49] introduced a streamlined semi-supervised framework employing linear sampling to dynamically assign class thresholds, thereby enhancing pseudo-label quality. Li et al. [50] devised a semi-supervised strategy under a GAN framework. It optimizes model performance through consistency self-training that learns joint distributions of labeled and unlabeled data. This method leverages pixel-level training labels to assess prediction confidence. Ma et al. [51] critically observed that existing consistency learning frameworks based on network perturbations suffer from excessive complexity. To address this, they developed a divergence network exploring lightweight multihead and multimodel perturbations to simultaneously enhance feature diversity and pseudo-label precision. However, these frameworks primarily focus on semi-supervised algorithmic shortcomings without considering the unique characteristics of RS images.

The second category focuses on enhancements targeting domain-specific challenges in RS. For instance, Gao et al. [52] developed a semi-supervised road extraction network via multiple consistency and multitask constraints (MCMCNet). It ensures the enhanced preservation of road continuity and trunk integrity. Chen et al. [53] proposed an adversarial learning-based semi-supervised network containing dual discriminators for road extraction from RS images. The two discriminators reinforce feature distribution alignment while maintaining consistency between labeled and unlabeled data. However, these semi-supervised methods typically focus on particular remote classes, such as roads and traffic signs, demonstrating limited generalization. For the more widely used multiclass RS images, Xin et al. [8] proposed a hybrid framework integrating consistency regularization and contrastive learning to address interclass similarities and intraclass variations in RS imagery. Nevertheless, their experiments predominantly focused on a 25% labeled data ratio, failing to demonstrate effectiveness under very few labeled data scenarios. Liu et al. [54] proposed a multilevel label-aware (MLLA) semi-supervised framework for RS scene classification to address interclass similarities. By extending semantic information extraction from unlabeled data from single level to multilevel, MLLA enhances scene classification performance. Zhang et al. [55] proposed the MDMASNet framework to tackle the rich multiscale information in RS images. MDMASNet employs a pyramid architecture that integrates deformable convolution and dilated convolution to extract features from objects of diverse shapes and sizes. Although these networks target rich multiscale information and high interclass similarities in semi-supervised RS, their proposed network models exhibit excessive complexity and fail to achieve nonintrusiveness. Huang et al. [56] considered the significant domain shift between different RS datasets. They proposed the semi-supervised bidirectional alignment for RS cross-domain scene classification. To address the inevitable issue of erroneous pseudo-labels and long-tail distribution in RS images, Huang et al. [9] introduced a decoupled weighting learning (DWL) framework. During training, the decoupled learning module separates the predictions of labeled and unlabeled data. And, the ranking weighting module attempts to adaptively weight each pseudo-label of the unlabeled data based on its relative confidence rank within the pseudo-class.

However, these RS semi-supervised models do not simultaneously address the rich multiscale information and high interclass similarities in RS images. In addition, some RS semi-supervised models do not provide open-source code,

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limiting their utility for subsequent work. This article is the first attempt to simultaneously address both challenges in the RS semi-supervised field and provide an open-source implementation. We propose the MUCA semi-supervised model to tackle these challenges and improve the performance of RS semi-supervised semantic segmentation.

III. Methods

This section is organized as follows. Section III-A describes the main optimization objectives, Section III-B introduces the principles of the MSUC module, and Section III-C presents the basic principles of the CTSA module. The overall structure of our approach is shown in Fig. 2.

A. Main Optimization Objectives

We define $D^L = \{(x_i^l, y_i)\}_{i=1}^{N_L}$ as labeled data and $D^U = \{(x_i^u)\}_{i=1}^{N_U}$ as unlabeled. Here $x_i^l \in \mathbb{R}^{H\times W\times 3}$ denotes the labeled image, $y_i \in \mathbb{R}^{H\times W\times K}$ is the ground truth of K classes, while $x_i^u \in \mathbb{R}^{H\times W\times 3}$ denotes the unlabeled image, N_L and N_U are the amount of labeled and unlabeled images. H and W specify the height and width of the image. It is worth noting that in general, N_U is much larger than N_L. For unlabeled images, weak augmentation (WA) and strong augmentation (SA) are performed to train the teacher and student networks, respectively. The main loss function is

$$\mathcal{L} = \mathcal{L}_S + \mathcal{L}_U = \frac{1}{N_L}\sum_{i=0}^{N_L}\mathcal{L}_{\mathrm{CE}}\left(p_i^l, y_i\right) + \mathcal{L}_U \tag{1}$$

where $\mathcal{L}_S$ represents the loss of supervised learning with labeled data, $\mathcal{L}_U$ represents the loss with unlabeled data, p_i^l is the prediction result of the labeled image x_i^l, and $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss. More specifically, for $\mathcal{L}_U$, it consists of three parts

$$\mathcal{L}_U = \mathcal{L}_C + \mathcal{L}_{\mathrm{MSUC}} + \mathcal{L}_{\mathrm{CTSA}}. \tag{2}$$

The first part $\mathcal{L}_C$ is the basic teacher-student consistency loss function, similar to the Mean teacher model [13]. The second part is the loss function $\mathcal{L}_{\mathrm{MSUC}}$ for MSUC, and the third part $\mathcal{L}_{\mathrm{CTSA}}$ is related to CTSA, which will be introduced later.

B. MSUC Module

Basic models of teacher-student architectures update the teacher's weights θ^t by the EMA with the student's weights θ^s. It integrates the information from different training steps. The teacher's weight θ_t^t is updated at training step t as $\theta_t^t = \alpha\theta_{t-1}^t + (1-\alpha)\theta_t^s$, where α is the EMA decay. This basic teacher-student architecture is widely used in semi-supervised networks. However, it is critical to take into account the inherent difference between RS and natural images. Objects in RS images exhibit a wide range of scales, spanning from large buildings to small vehicles, and the same object can appear at varying sizes under different resolutions. Therefore, semi-supervised learning in RS images is difficult to extract effective features. To solve this problem, we propose an MSUC module for the student to progressively learn multiscale features from reliable targets, as shown in Fig. 3. Given a batch of training images, the teacher model not only generates pseudo-labels for target prediction but also estimates the uncertainty of each target at multiple feature levels. It optimizes the student model through multilevel consistency loss. This will motivate the model to learn different layers and sizes of features from RS images.

1) Uncertainty Estimation: We first compute pseudo-label uncertainty to mitigate the impact of low-quality pseudo-labels in semi-supervised learning frameworks. This is to enable the model to progressively learn more reliable targets from multiscale information. Specifically, we use Monte Carlo dropout [57] to estimate uncertainty. The Monte Carlo method can not only calculate the uncertainty between different classes but also evaluate the uncertainty in the multiple predictions of a neural network. The basic principle of Monte Carlo dropout is to make predictions of T times for the same sample with the same model. These T predictions are different, and their variance is calculated to compute the model uncertainty. In detail, we perform T random forward passes for the teacher model, each of which uses the random dropout and noise. Thus, for each pixel in the image, we obtain a set of softmax probability vectors: $\{p_t\}_{t=1}^T$. We choose the prediction entropy as a measure of the approximate uncertainty because it has a fixed range. Formally, the prediction entropy can be summarized as

$$u_k = \frac{1}{T}\sum_{t=1}^{T} p_t^k \tag{3}$$

$$u = -\sum_{k=1}^{K} u_k \log\left(u_k\right) \tag{4}$$

where p_t^k is the result of pixels of the class k in the tth prediction procedure. In this way, the uncertainty of multiple predictions for the same class can be calculated. According to (3) and (4), the value u increases when the network provides completely opposite predictions many times. Moreover, (3) and (4) can also take into account the uncertainty between different classes. If the neural network gives similar predictions across multiple classes, the uncertainty value u will also increase.

2) Consistency Loss Functions for Multiscale Uncertainty: To formally describe MSUC, we denote the encoded features from the four stages of the encoder as $V_i \in \mathbb{R}^{C_i\times H_i\times W_i}$, $i \in \{1, 2, 3, 4\}$, where C_i, H_i, and W_i denote the number of channels, the height, and the width of the feature maps, respectively, from the ith stage, as shown in Fig. 3. More specifically, V_1 is the output of the first stage with the lowest abstraction but the highest spatial resolution, and V_4 is the output of the fourth stage with the highest abstraction but the lowest spatial resolution. The final consistency loss function was computed by aligning the encoded visual features at each stage of the student model and the teacher model

$$\mathcal{L}_{\mathrm{MSUC}} = \sum_{i=1}^{4}\frac{\sum_m \mathbb{I}\left(u_{i_m} < H\right)\mathcal{L}_\varrho\left(V_{i_m}^t - V_{i_m}^s\right)}{\sum_m \mathbb{I}\left(u_{i_m} < H\right)} \tag{5}$$

where $V_{i_m}^t$ and $V_{i_s}^s$ are the outputs of the teacher and student encoders, respectively, at the pixel m of the ith stage, u_{i_m}

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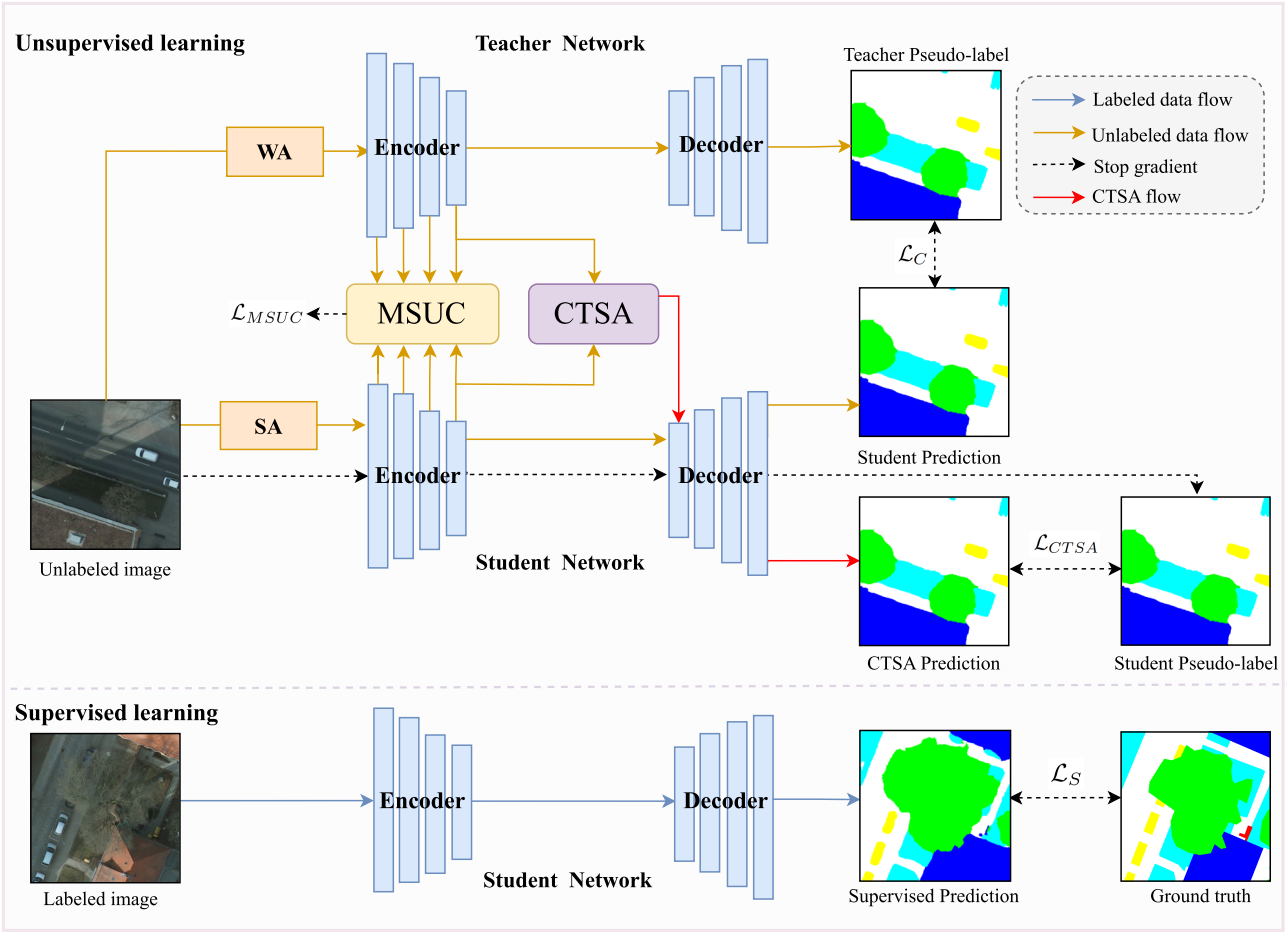


Fig. 2. Overall structure of our semi-supervised semantic segmentation model for RS images. The top and bottom sections correspond to the unsupervised learning for unlabeled data and supervised learning for labeled data, respectively. Blue lines indicate labeled data flow, yellow lines indicate unlabeled data flow, red lines indicate CTSA flow, and dashed lines indicate data flow where gradient updating stops. The teacher network and student network interaction part contains the MSUC module and CTSA module proposed in this article.

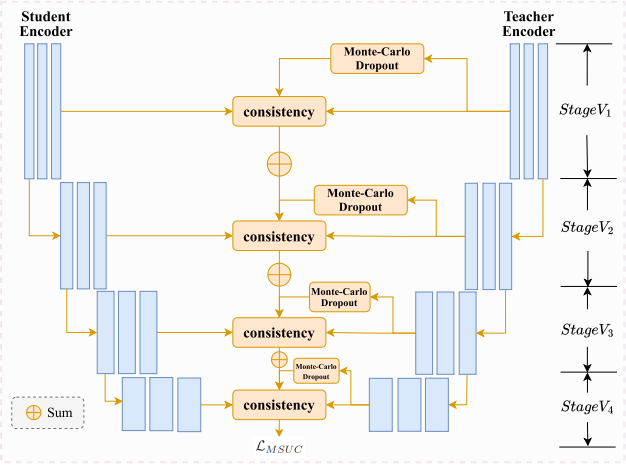


Fig. 3. MSUC model structure. Uncertainty is computed for the feature layer at each stage of the teacher and student networks, and features with high uncertainty are discarded when computing consistency regularization.

is the uncertainty at the pixel m of the ith stage, $\mathbb{I}$ is the indicator function, and H is the threshold. Specifically, for threshold setting, we adopt a dynamic threshold approach using a Gaussian ramp-up curve [58]. During the training process, the uncertainty threshold H is gradually increased from the initial $(1/2)H_{\max}$ to $H_{\max}$, where $H_{\max}$ denotes the maximum uncertainty threshold setted as ln2 in our experiments. Consequently, as training progresses, our method filters out progressively fewer samples, enabling the student model to gradually learn from relatively certain to more uncertain cases. This design ensures model stability in early training phases by eliminating substantial low-quality pseudo-label pixels. As the training progresses, the model becomes gradually more reliable, allowing more samples to be selected in the later stages of training, even if their uncertainty is slightly higher. Our $H_{\max}$ setting still filters out pseudo-label pixels with higher uncertainty. $\mathcal{L}_\varrho$ is the Huber loss function, which is formulated as follows:

$$\mathcal{L}_\varrho\left(V_{i_m}^t - V_{i_m}^s\right) = \begin{cases} \frac{1}{2}\left(V_{i_m}^t - V_{i_m}^s\right)^2, & \text{if } |V_{i_m}^t - V_{i_m}^s| \le \varrho \\ \varrho|V_{i_m}^t - V_{i_m}^s| - \frac{1}{2}\varrho^2, & \text{otherwise} \end{cases} \tag{6}$$

where ϱ is the soft threshold for Huber loss, which is set to 1.0 in this article.

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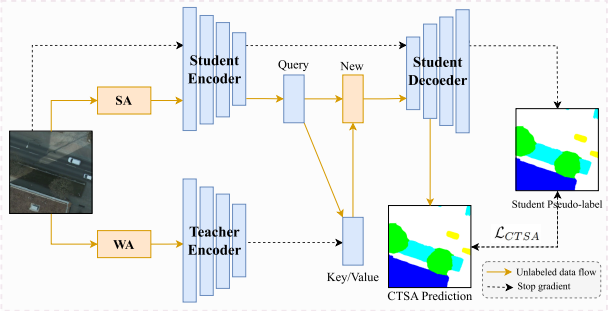


Fig. 4. CTSA model structure. An attention approach allows the teacher network to guide the student network in reconstructing the student network's encoder results.

C. Cross-Teacher-Student Attention

We propose the CTSA model, as illustrated in Fig. 4, to promote the ability of a semi-supervised segmentation method for objects with high interclass similarities.

Consistency regularization methods are based on the assumption of smoothness [59]. This means that a robust model should produce similar predictions for points and their variants with noise. In other words, a model trained with consistency methods should not be affected by different perturbations added to the data. However, such semi-supervised frameworks ignore the inherent domain gap between natural and RS images. One RS image contains much richer content than one natural image, and objects of different categories are commonly intertwined. Therefore, perturbations will make the RS image contain more confusing information. The features learned by the neural network from different perturbed RS images may lead to some outliers as pseudo-annotations, misleading the training of the model on unannotated data.

To address this problem, we use the features extracted by the student and teacher networks via different perturbations to reconstruct the features of the unlabeled data. We use the encoder result of the student as query and the encoder result of the teacher as key and value. The teacher guides the reconstruction of the encoder of the student, as shown in Fig. 4. Specifically, we calculate the similarity between each channel of the unlabeled features of the student and teacher networks. The channels with higher similarity are more important in reconstructing the unlabeled features. This ensures enhanced feature extraction within the same category while reducing cross-category feature weights.

More importantly, the new features constructed through CTSA effectively integrate the advantages of the teacher and student network encoders. Specifically, the SA data fed into the student encoder provide the original features, which are diverse and rich. Meanwhile, the teacher encoder, receiving WA data, reconstructs another set of smooth and stable features. The decoder is trained using features generated from strongly supervised data along with features reconstructed by CTSA, resulting in better optimization outcomes. Particularly in the early stages of semi-supervised learning, where training instability and low-quality pseudo-labels often hinder efficient convergence, the teacher network encoder provides more easily learnable features for the student network decoder. This improves the model's stability, mitigates training fluctuations, accelerates convergence, and enhances training efficiency.

Algorithm 1 Training Process of MUCA

Input:
$D_U^e = \{(x_i^u)\}_{i=1}^{N_U^e}$, $D_L^e = \{(x_i^l, y_i)\}_{i=1}^{N_L^e}$
Outpt:
Θ: optimal model parameters
while until converge:
  for x_i^l, x_i^u in D_L^e, D_U^e:
    $\mathcal{L}_S = CE(model_s(x_i^l), y_i)$
    $W_x_i^u = WeakAugment(x_i^u)$
    $S_x_i^u = StrongAugment(x_i^u)$
    $V^s, Enc_s, Pre_s = model_s(S_x_i^u)$
    $V^t, Enc_t, Pseudo_t = model_t(W_x_i^u)$
    $\mathcal{L}_C = Huber(Pre_s, Pseudo_t)$
    for j in [1:4]:
      Calculate uncertainty u via Eq. (4)
      if $u < H$:
        Calculate $\mathcal{L}_{MSUC}$ of j stage via Eq. (5)
        $\mathcal{L}_{MSUC} += \mathcal{L}_{MSUC}$
    end for
    $Pseudo_s = model_s(x_i^u).detach()$
    $Pre_{ctsa} = model_s_d(CTSA(Enc_s, Enc_t))$
    $\mathcal{L}_{CTSA} = CE(Pre_{ctsa}, Pseudo_s)$
    $\mathcal{L}_U = \mathcal{L}_C + \mathcal{L}_{CTSA} + \mathcal{L}_{MSUC}$
  end for
  $\mathcal{L} = \mathcal{L}_S + \mathcal{L}_U$
  Back propagating the $\mathcal{L}$ and update Θ
  Mark the current model checkpoint and use it to predict pseudo labels of unlabeled images
  Save the best checkpoint Θ
return Θ

We further clarify the CTSA formulation as follows. Given the student and teacher characteristics V_4^s and V_4^t after the fourth stage of the encoder, we set V_4^s as query and V_4^t as key and value, in a multihead manner

$$F^{V_4^s} = \text{flatten}\left(V_4^s\right), \quad F^{V_4^t} = \text{flatten}\left(V_4^t\right) \tag{7}$$

$$q = F^{V_4^s} w_q, \quad k = F^{V_4^t} w_k, \quad v = F^{V_4^t} w_v \tag{8}$$

where $w_q, w_k, w_v \in \mathbb{R}^{C\times 2C}$ are the Transformer weights, $F^{V_4^s}, F^{V_4^t} \in \mathbb{R}^{C\times d}$, d is the number of patches, and C is the channel dimension. The CTSA is defined as

$$V_{\text{out}} = \phi\left[\psi(q^{\mathrm{T}}k)v^{\mathrm{T}}w_{\text{out}}\right] \tag{9}$$

where ψ denotes instance normalization, and ϕ is the softmax function. V_{out} is fed into the decoder to get the final predicted value of the CTSA module as described in the following equation:

$$p^u = \text{Decoder}(V_{\text{out}}). \tag{10}$$

The loss function for the CTSA module is calculated as (11), where $\hat{y}_i$ is the generated pseudo-label from unlabeled data

$$\mathcal{L}_{\text{CTSA}} = \frac{1}{N_U}\sum_{i=0}^{N_U} \mathcal{L}_{\text{CE}}\left(p_i^u, \hat{y}_i\right). \tag{11}$$

Algorithm 1 gives the core pseudocode for the training process of MUCA on the unlabeled data. D_U^e and D_L^e are the

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unsupervised and supervised datasets for one epoch. N_U^e and N_L^e are the number of images in D_U^e and D_L^e. model_s and model_t denote the student and teacher models. model_s_d is the decoder of the student model. V^s and V^t are the feature maps extracted by the encoders of the student network and the teacher network. Enc_s and Enc_t are the output results of the student network and teacher network encoders. Pre_s represents the final prediction of the student network. Pseudo_t represents the pseudo-labels generated by the teacher network, and Pseudo_s represents the pseudo-labels generated by the student network. CE is the cross-entropy loss function, and Huber is the Huber loss function

IV. Experiment

In this section, we conduct experiments on semi-supervised semantic segmentation of RS images to evaluate the proposed MUCA model. We first perform ablation studies on MUCA to validate the effectiveness of the proposed modules. Then, we conducted comparative experiments to compare MUCA with several SOTA models. Our code is released at https://github.com/wangshanwen001/RS-MUCA

A. RS Dataset

1) LoveDA: The LoveDA RS dataset [60] consists of 5987 images and 166 768 annotated objects from three different cities. Each image is 1024 × 1024 with a spatial resolution of 0.3 m. The dataset has seven segmentable classes, such as Building, Road, Water, Barren, Forest, Agriculture, and Background. Due to the memory limitations of the GPU, each image in the LoveDA dataset is resized and cropped to 512 × 512, producing a total of 16 764 cropped images for subsequent deep learning training. During training, we divide the dataset into a training set, a validation set, and a testing set by a ratio of 6:2:2.

2) ISPRS-Potsdam: The ISPRS-Potsdam RS dataset is provided to facilitate research on RS images semantic segmentation [61]. This dataset has a resolution of 0.05 m and consists of 38 super-large satellite RS images of 6000 × 6000 size. There are six segmentable classes in the dataset, including Impervious surfaces, Building, Low vegetation, Tree, Car, and Background. To facilitate the training procedure, we crop the original image into 512 × 512, and the number of cropped images is 5472. The dataset is divided into training, validation, and testing sets by a ratio of 6:2:2.

B. Data Augmentation and Experiment Details

We first augment the labeled images via geometric transformations (i.e., image scaling, horizontal flipping, vertical flipping, and length and width warping) and Gaussian blurring. The WA used for unlabeled images is geometric augmentation, and SA is done by methods such as CutMix [62].

The experiments are conducted on A6000 (48G) with Python-3.8.10, PyTorch-1.13.0, and CUDA v11.7. The optimization algorithm is stochastic gradient descent with an initial learning rate of 0.007. The learning rate is updated on each epoch, the weight decay is set to 0.0001, and the minimum learning rate is set to 0.00007. Then, we train the semi-supervised segmentation model using 1%, 5%, and 10% of the labeled images and the remaining unlabeled images.

C. Evaluation Metrics

Most of the popular semantic segmentation metrics are employed to comprehensively evaluate the performance, e.g., intersection and concurrence ratio (IoU), $F1$-Score ($F1$), and Cohen's Kappa, as shown in the following equations:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$F1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{15}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{16}$$

$$\text{PRE} = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{TN} + \text{FN})(\text{TN} + \text{FP})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} \tag{17}$$

$$\text{Kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}} \tag{18}$$

where TP is the number of correctly predicted positive pixels, TN is the number of correctly predicted negative pixels, FP is the number of incorrectly predicted positive pixels, and FN is the number of incorrectly predicted negative pixels. In our experiments we average IoU and $F1$ across all classes, so the final experimental metrics also include mean IoU (mIoU) and mean F1 (mF1). Then, mIoU and mF1 is calculated as follows:

$$\text{mIoU} = \frac{\sum_{i=1}^{K} \text{IoU}_i}{K} \tag{19}$$

$$\text{mF1} = \frac{\sum_{i=1}^{K} F1_i}{K} \tag{20}$$

where IoU_i means the IoU of the ith class, and K is the number of classes.

D. Ablation Study

1) Ablation of Components: We ablate each component step-by-step to investigate their performance. It is worth noting that we also evaluate the multiscale consistency loss function of the MSUC module without uncertainty, which is denoted as NoUC. Differing from the MSUC module, NoUC only uses standard consistency regularization to compute the loss for each stage feature map. The ablation studies are conducted on the ISPRS-Potsdam dataset with a labeled data ratio of 5%. The baseline is the supervised model with 5% labeled data. The semantic segmentation model used in this section is SegFormer-B2.

The experimental results are shown in Table I. NoUC, which despite solely utilizes standard consistency regularization to compute the loss function at each stage, still improves semantic segmentation performance with limited labeled data. Compared with NoUC, MSUC achieves improved mIoU on both the ISPRS-Potsdam dataset and the LoveDA dataset.

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TABLE I
ABLATION STUDY ON THE EFFECTIVENESS OF COMPONENTS

Dataset	NoUC	MSUC	CTSA	mIoU (5%)
ISPRS-Potsdam				72.01
	✓			73.14
		✓		73.95
			✓	73.80
		✓	✓	74.62
LoveDA				44.20
	✓			48.21
		✓		50.02
			✓	49.81
		✓	✓	50.97

TABLE II
IMPACT OF CTSA MODULE IN THE TEST STAGE

Dataset	Stage	CTSA	mIoU (5%)
ISPRS-Potsdam	Test		73.80
		✓	73.87
LoveDA	Test		49.81
		✓	49.85

TABLE III
ANALYSIS OF MULTILEVEL FEATURE MAPS OF MSUC MODULE

Dataset	V_1	V_2	V_3	V_4	mIoU (5%)
ISPRS-Potsdam					72.01
	✓				72.41
		✓			72.55
			✓		72.76
				✓	72.96
LoveDA					44.20
	✓				45.97
		✓			46.72
			✓		47.09
				✓	47.91

TABLE IV
ANALYSIS OF DEEP AND LOW FEATURES FUSION FOR MSUC MODULE

Dataset	V_1	V_2	V_3	V_4	mIoU (5%)
ISPRS-Potsdam					72.01
				✓	72.96
			✓	✓	73.42
		✓	✓	✓	73.66
	✓	✓	✓	✓	73.95
LoveDA					44.20
				✓	47.91
			✓	✓	49.10
		✓	✓	✓	49.64
	✓	✓	✓	✓	50.02

These results indicate that uncertainty computation enables the student network to discard unreliable information. The model integrated with the CTSA module demonstrates a significant mIoU improvement over the baseline. This demonstrates that the CTSA module effectively enhances the segmentation performance of a semi-supervised model on RS imagery.

Should We Integrate the CTSA Module in the Test Stage? The CTSA module requires both teacher-student networks to complete the attention. However, generally speaking, we only save one final checkpoint file. In this study, we save both the teacher and student checkpoint files in each epoch. We then investigate the influences of CTSA on the mIoU in the testing stage. As shown in Table II, the impact of the CTSA module is almost negligible in the testing phase. The double-disk consumption and limited performance improvement make us abandon the CTSA module in the testing phase.

2) Analysis of Multilevel Feature Maps of MSUC Module: MSUC encodes visual features from the four stages of the encoder, i.e., $V_i \in \mathbb{R}^{C_i \times H_i \times W_i}$, $i \in \{1, 2, 3, 4\}$. Therefore, it is critical to explore the impact of multilevel (i.e., low to high) deep features on our MSUC. From Table III, we can see that the deep MSUC feature V_4 achieves greater improvement than the other low-level features, while the lowest feature V_1 makes the worst mIoU. The reason is that when only utilizing features extracted from a single stage of the model, deep features are more effective than low-level features on forming global semantic representations such as object categories and overall contours.

3) Analysis of Features Fusion for MSUC Module: We fuse deep and low features stage by stage for the MSUC module. Specifically, we first employ the deep feature V_4 alone, and then gradually add all the other level features. Table IV shows the results of these different combinations. From the results, we observe that the mIoU gradually increases as lower level features are added stage by stage. This is because low-level features, generated by the shallow stages of the network, have small receptive fields and primarily capture local details such as image edges and textures. Deep features accumulate large-scale contextual information through repeated downsampling and convolutional operations, forming global semantic representations like object categories and overall contours. Therefore, the high resolution and detail retention capability of low-level features make them inherently suited for fine-grained analysis. While the lower resolution and broader receptive fields of deep features enable them to excel in large-scale semantic comprehension. The fusion of deep and low features improves the consistency of the semi-supervised model at different stages and scales, significantly improving the model's capability to capture multiscale characteristics of target objects. Consequently, it optimizes semantic segmentation performance in RS images.

4) Confusion Matrix Visualization: We visualized the confusion matrices of the ISPRS-Potsdam dataset at a labeled data ratio of 5%, as shown in Figs. 5 and 6. It intuitively demonstrates the improvement of the semi-supervised semantic segmentation with MUCA. We perform a row normalization on the confusion matrix, where the redder of the color, the closer to the truth. We can see that the introduced MUCA significantly improves the ability to distinguish hard-to-differentiate classes, such as Building, Impervious surfaces, Low vegetation, and Tree, on the ISPRS-Potsdam dataset. Furthermore, it demonstrates significant enhancement in segmenting Barren, Agriculture, Building, and Background classes on the LoveDA dataset.

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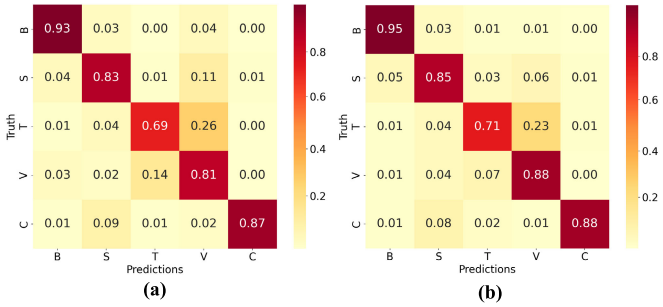


Fig. 5. Visualization confusion matrix of model (a) without MUCA and (b) with MUCA on the ISPRS-Potsdam dataset. Car (C), Low vegetation (V), Tree (T), Impervious surfaces (S), and Building (B).

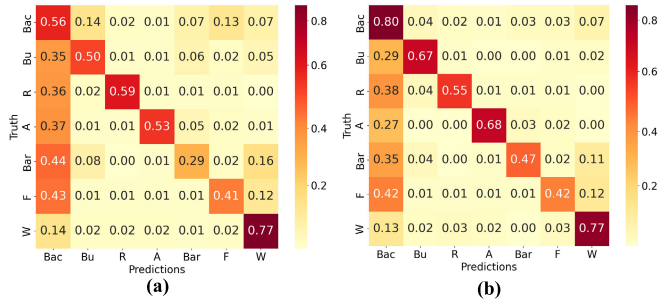


Fig. 6. Visualization confusion matrix of model (a) without MUCA and (b) with MUCA on LoveDA dataset. Background (Bac), Building (Bu), Road (R), Agriculture (A), Barren (Bar), Forest (F), and Water (W).

E. Comparison With SOTA Methods on RS Datasets

This section conducts experiments on ISPRS-Potsdam and LoveDA datasets, compared with the SOTA methods, including Mean teacher [10], CutMix [62], CCT [43], CPS [44], LSST [49], Fixmatch [11], Unimatch [18], DWL [9], and Allspark [12]. Specifically, we show the results for labeled data ratios of 1%, 5%, and 10% with SegFormer-B2 as the segmentation model to verify the effectiveness, respectively. We also conduct experiments of OnlySup and FullySup as reference, where OnlySup uses only 1%, 5%, and 10% of labeled data for training, while FullySup employs the entire dataset, which represents the 100% labeled data. The experimental results are presented in Tables V and VI, respectively. From the viewpoint of methods, our observations reveal the following trends. Methods such as CutMix and CCT exhibit a performance decline when incorporating unlabeled data into the training process, compared with the basic OnlySup method. In contrast, methods like FixMatch, UniMatch, DWL, and the proposed MUCA demonstrate a beneficial impact on performance. MUCA achieves the best results. There are many possible reasons. For example, methods like CutMix and CCT use all unlabeled data for consistency regularization during training without selective filtering. This may inadvertently result in memorizing a large number of incorrect pseudo-labels. On the other hand, FixMatch, UniMatch, DWL, and MUCA employ strategies, such as careful filtering or weighting of pseudo-labels in the unlabeled data, which reduces the harmful impact of low-confidence samples and prevents the dominance of most classes.

In addition, an interesting phenomenon is that sometimes the performance of UniMatch is lower than that of FixMatch. UniMatch extends FixMatch by introducing an additional perturbation branch and a feature reperturbation branch to enrich the representation space. Both branches are supervised by a weakly augmented branch. However, RS datasets often contain too many hard-to-distinguish classes and noise. They hinder the training process on the unlabeled data, which could be exacerbated by the inclusion of additional pseudo-supervision branches. In contrast, our method achieves better results by using MSUC to exclude pseudo-labels with high uncertainty and utilizing CTSA to enhance the correlation between WA and SA.

Furthermore, it is worth noting that our model outperforms all other SOTA methods on ISPRS-Potsdam and LoveDA datasets when the labeled data ratios are 1%, 5%, and 10%, respectively. This is a good demonstration of the fact that our semi-supervised model achieves better applicability on RS datasets with a very small labeled data ratio.

For specific classes, our model performs well from expansive classes to small classes, such as large-scale Building and small-scale Car. Our model achieves SOTA performance on Building, which is a class with extreme scale variations. Both datasets include categories that are difficult to distinguish. For example, the ISPRS-Potsdam dataset includes Building, Low vegetation, Tree, and Impervious surfaces, while the LoveDA dataset includes categories such as Building, Barren, Forest, and Agriculture. More specifically, classes including Building, Impervious surfaces, Barren, Road, and Background predominantly exhibit grayish tones and may share similar rough or smooth textures. Meanwhile, Low vegetation, Tree, Forest, and Agriculture, which predominantly appear in green hues, also pose segmentation challenges for models. Our method, MUCA, achieves an improvement on IoU for almost all classes compared with OnlySup when the labeled data ratios are 1%, 5%, and 10%. In addition, when the labeled data ratios are 1%, 5%, and 10%, MUCA outperforms other SOTA models in IoU performance for the Building, Car, and Impervious surfaces classes in the ISPRS-Potsdam dataset and for the Background, Building, Barren, and Agriculture classes in the LoveDA dataset. Although MUCA does not achieve the best IoU for all classes, this phenomenon is understandable and explainable. This is because different models have different focuses, which make one algorithm perform exceptionally well on specific classes while underperforming for others. However, focusing solely on improving a single class is not the research focus of this article. For example, while the DWL model achieves significant improvement in the Low vegetation class, its performance on the Building, Car, and Impervious surfaces classes is unsatisfactory.

F. Model Generalizability Experiments

The proposed model is nonintrusive and can be easily integrated into existing semantic segmentation networks without changing the network structure itself. Therefore, we conduct experiments by integrating our model into several popular networks, including U-Net, PSPNet, DeepLabv3+, and SegFormer-B2 on RS datasets. These experiments are designed

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TABLE V
COMPARISON RESULTS WITH SOTA METHODS ON ISPRS-POTSDAM DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. IOU, MIOU, MF1, AND KAPPA ARE REPRESENTED AS PERCENTAGES

Ratio	Model	IoU					mIoU/mF1/Kappa
		Building	Low vegetation	Tree	Car	Impervious surfaces	
1%	OnlySup	69.80	59.39	59.50	66.50	55.85	62.21 / 76.58 / 0.6531
	Mean teacher [10]	72.53	58.98	60.43	67.97	67.39	65.46 / 79.01 / 0.7187
	CutMix [62]	55.58	42.05	49.72	50.86	39.40	47.52 / 64.21 / 0.5121
	CCT [43]	54.48	61.28	48.56	52.95	60.71	55.59 / 71.34 / 0.5761
	CPS [44]	59.35	69.16	62.89	59.88	66.33	63.52 / 77.63 / 0.6539
	LSST [49]	68.74	75.24	54.74	62.09	68.80	65.92 / 79.25 / 0.6847
	FixMatch [11]	76.95	71.59	64.71	65.85	72.81	70.38 / 82.53 / 0.7287
	UniMatch [18]	76.52	70.99	**65.44**	66.62	72.64	70.44 / 82.59 / 0.7291
	DWL [9]	72.34	**77.08**	62.74	62.57	72.22	69.39 / 81.79 / 0.7191
	AllSpark [12]	83.70	65.92	59.64	69.77	75.31	70.87 / 82.68 / 0.7827
	Our (MUCA)	**84.56**	66.98	56.96	**71.52**	**76.64**	**71.33 / 82.92 / 0.7880**
5%	OnlySup	84.22	64.10	60.81	74.04	76.88	72.01 / 83.10 / 0.7371
	Mean teacher [10]	82.15	65.92	67.11	72.21	74.60	72.40 / 83.86 / 0.7403
	CutMix [62]	52.94	68.86	41.51	58.33	54.82	55.29 / 70.79 / 0.5783
	CCT [43]	72.90	80.25	64.23	58.32	74.42	70.02 / 82.12 / 0.7236
	CPS [44]	76.53	84.34	57.98	69.45	75.39	72.74 / 83.78 / 0.7492
	LSST [49]	69.26	84.55	67.33	67.49	73.86	72.50 / 83.67 / 0.7399
	FixMatch [11]	78.12	74.87	**68.89**	66.58	75.30	72.75 / 84.15 / 0.7497
	UniMatch [18]	78.24	73.59	67.17	66.64	75.07	72.14 / 83.73 / 0.7432
	DWL [9]	74.81	**85.64**	66.38	62.99	75.68	73.10 / 84.22 / 0.7507
	AllSpark [12]	85.57	67.62	60.61	73.48	77.15	72.88 / 84.04 / 0.7989
	Our (MUCA)	**88.45**	69.53	61.39	**74.18**	**79.56**	**74.62 / 85.15 / 0.8166**
10%	OnlySup	83.01	69.26	66.83	75.02	75.47	73.92 / 84.89 / 0.7564
	Mean teacher [10]	84.76	69.28	68.83	71.66	76.51	74.21 / 85.07 / 0.7578
	CutMix [62]	64.55	80.99	64.79	65.50	68.01	68.77 / 81.34 / 0.7109
	CCT [43]	73.09	83.94	61.12	60.45	73.06	70.33 / 82.27 / 0.7265
	CPS [44]	77.80	87.15	61.12	68.48	75.89	74.09 / 84.55 / 0.7533
	LSST [49]	70.92	86.06	68.91	70.22	74.89	74.20 / 84.95 / 0.7549
	FixMatch [11]	77.97	76.17	70.09	70.97	76.14	74.27 / 85.20 / 0.7606
	UniMatch [18]	77.34	87.75	**70.79**	56.65	76.46	73.80 / 84.52 / 0.7599
	DWL [9]	76.37	**88.42**	66.54	64.37	77.14	74.57 / 85.16 / 0.7628
	AllSpark [12]	86.29	69.83	64.17	75.23	78.31	74.76 / 85.35 / 0.8144
	Our (MUCA)	**88.02**	70.58	64.53	**75.20**	**79.92**	**75.65 / 85.90 / 0.8245**
100%	FullySup	90.81	72.08	68.32	76.09	82.56	77.97 / 87.40 / 0.8459

to evaluate the general applicability of the MUCA on segmentation models. The first three of these models are classical CNN semantic segmentation models, and the last one is based on the Transformer architecture. We comparatively analyze the performance improvement of MUCA with Onlysup and NoUC on these models. NoUC, as clearly defined in Section IV-D, refers to a simplified method that does not perform multiscale uncertainty estimation and does not incorporate the CTSA module. Instead, it relies solely on standard consistency regularization to calculate the loss of feature maps at each stage. The model's general applicability experiments were performed on 5% labeled training data, and the results are shown in Table VII.

It can be seen that the semi-supervised model MUCA achieves significant improvements with 5% labeled training data. It enhances the performance of popular semantic segmentation models, including U-Net, PSPNet, DeepLabv3+, and SegFormer-B2. This shows that our MUCA has the ability to improve the performance of classical CNN-based semantic segmentation models and novel Transformer architecture models.

G. Visualization Comparison

We conducted visual comparison experiments to see the advantages of our approach more intuitively and clearly. Figs. 7 and 8 show the visual comparison of several

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TABLE VI
COMPARISON RESULTS WITH OTHER SOTA METHODS ON THE LOVEDA DATASET. THE BEST RESULTS ARE IN BOLD. IOU, MIOU, MF1, AND KAPPA ARE REPRESENTED AS PERCENTAGES

Ratio	Model	IoU							mIoU/mF1/Kappa
		Background	Building	Road	Water	Barren	Forest	Agriculture	
1%	OnlySup	41.96	37.86	36.74	57.03	9.02	31.70	37.73	36.01 / 51.41 / 0.4141
	Mean teacher [10]	44.73	42.53	40.34	50.92	11.37	26.88	54.40	38.73 / 52.81 / 0.4251
	CutMix [62]	36.04	24.69	10.03	24.60	3.43	6.67	10.19	16.52 / 26.85 / 0.1362
	CCT [43]	37.16	22.41	27.86	43.98	14.51	25.38	36.67	29.71 / 44.99 / 0.3517
	CPS [44]	46.52	20.87	27.85	50.55	0.01	33.16	34.60	30.51 / 44.28 / 0.3802
	LSST [49]	44.73	41.90	39.90	62.65	29.27	31.26	48.29	42.57 / 59.00 / 0.4817
	FixMatch [11]	46.78	51.20	50.21	67.27	11.53	**36.79**	50.26	44.86 / 60.02 / 0.5175
	UniMatch [18]	46.53	51.38	49.36	**67.74**	10.86	33.40	52.28	44.51 / 59.51 / 0.5175
	DWL [9]	48.74	56.79	**51.59**	63.42	22.56	35.20	55.38	47.67 / 63.40 / 0.5534
	AllSpark [12]	63.87	47.70	46.05	61.52	35.31	30.94	55.64	48.72 / 63.29 / 0.5502
	Our (MUCA)	**64.89**	**56.03**	47.14	63.86	**35.81**	22.57	**58.18**	**49.78 / 63.48 / 0.5753**
5%	OnlySup	48.00	43.34	50.56	61.42	23.14	**37.63**	45.00	44.20 / 60.46 / 0.5061
	Mean teacher [10]	49.73	46.22	42.34	60.93	31.51	35.79	44.22	44.39 / 61.81 / 0.5151
	CutMix [62]	41.48	41.62	38.77	47.44	14.69	28.09	31.05	34.73 / 50.65 / 0.3692
	CCT [43]	46.80	44.62	46.80	60.95	24.83	29.03	44.30	42.48 / 58.74 / 0.4850
	CPS [44]	48.90	49.64	47.97	60.27	4.67	36.09	47.32	42.12 / 56.90 / 0.4976
	LSST [49]	51.48	45.66	52.66	67.63	33.52	35.80	48.60	47.91 / 64.10 / 0.5434
	FixMatch [11]	45.40	53.05	**51.22**	66.73	28.53	27.25	54.30	44.64 / 62.45 / 0.5378
	UniMatch [18]	50.20	54.49	50.46	67.18	26.79	30.06	54.86	47.72 / 63.46 / 0.5543
	DWL [9]	48.75	55.00	51.53	**69.49**	29.46	36.59	52.11	48.99 / 64.88 / 0.5597
	AllSpark [12]	65.09	55.06	47.59	67.10	34.67	26.86	51.87	49.75 / 64.91 / 0.5682
	Our (MUCA)	**67.29**	**56.04**	48.37	61.02	**36.21**	30.76	**57.09**	**50.97 / 64.92 / 0.5856**
10%	OnlySup	47.02	51.04	50.73	59.42	31.66	38.25	51.26	47.05 / 63.52 / 0.5254
	Mean teacher [10]	50.45	55.75	43.56	66.15	35.24	36.96	45.64	47.68 / 64.18 / 0.5387
	CutMix [62]	46.73	49.60	47.36	59.99	29.06	37.77	40.60	44.44 / 60.99 / 0.4837
	CCT [43]	44.07	45.22	47.65	57.12	24.41	32.50	45.07	42.29 / 58.73 / 0.4762
	CPS [44]	51.30	54.93	52.57	53.37	18.39	37.59	53.24	45.91 / 61.78 / 0.5479
	LSST [49]	50.69	49.50	52.63	69.85	27.25	36.24	52.06	48.32 / 64.17 / 0.5565
	FixMatch [11]	52.02	55.59	53.20	57.91	25.86	40.83	57.50	48.99 / 64.97 / 0.5676
	UniMatch [18]	51.80	53.95	51.17	58.15	25.60	38.72	54.86	47.75 / 63.86 / 0.5639
	DWL [9]	49.94	56.66	**53.89**	**70.35**	30.62	**41.49**	53.13	50.87 / 66.64 / 0.5753
	AllSpark [12]	67.13	56.16	40.67	63.58	32.54	32.03	56.91	49.86 / 63.97 / 0.5751
	Our (MUCA)	**68.69**	**58.20**	41.82	65.62	**37.09**	35.01	**57.38**	**51.97 / 66.72 / 0.5901**
100%	FullySup	68.84	58.57	48.02	70.39	43.28	38.59	62.30	55.71 / 69.10 / 0.6291

semi-supervised semantic segmentation methods on the ISPRS-Potsdam dataset and the LoveDA dataset. In addition, we have highlighted the mis-segmented areas of other SOTA methods with dashed ellipses, while emphasizing the comparatively accurate segmentation results of our approach in the same regions for contrast.

For the ISPRS-Potsdam dataset, the Fixmatch and Unimatch make errors in segmenting regions for the Low vegetation, Tree, Impervious surfaces, and Building classes. Allspark and DWL successfully recognize part of the region where Low vegetation classes are mixed with Tree classes, but the segmentation results for the Building class exhibit inaccuracies. In contrast, our method shows significant advantages in accurately recognizing and segmenting Building (blue), Car (yellow), and Tree (green). Especially, we can see that methods, including Fixmatch and Unimatch, show large-scale segmentation errors when Car, Low vegetation, and Tree are mixed. In addition, we overlap our segmentation results with the original image in the last column to clearly show our advantage. This overlapped image demonstrates that our model achieves excellent visual results, particularly in the Tree and Car categories.

The objective metrics of all models on the LoveDA dataset are lower compared with those on the ISPRS-Potsdam dataset. Consequently, the visualization effects of all models become less effective compared with their performance on the ISPRS-Potsdam dataset. Specifically, FixMatch and UniMatch exhibit noticeable errors in the segmentation areas of

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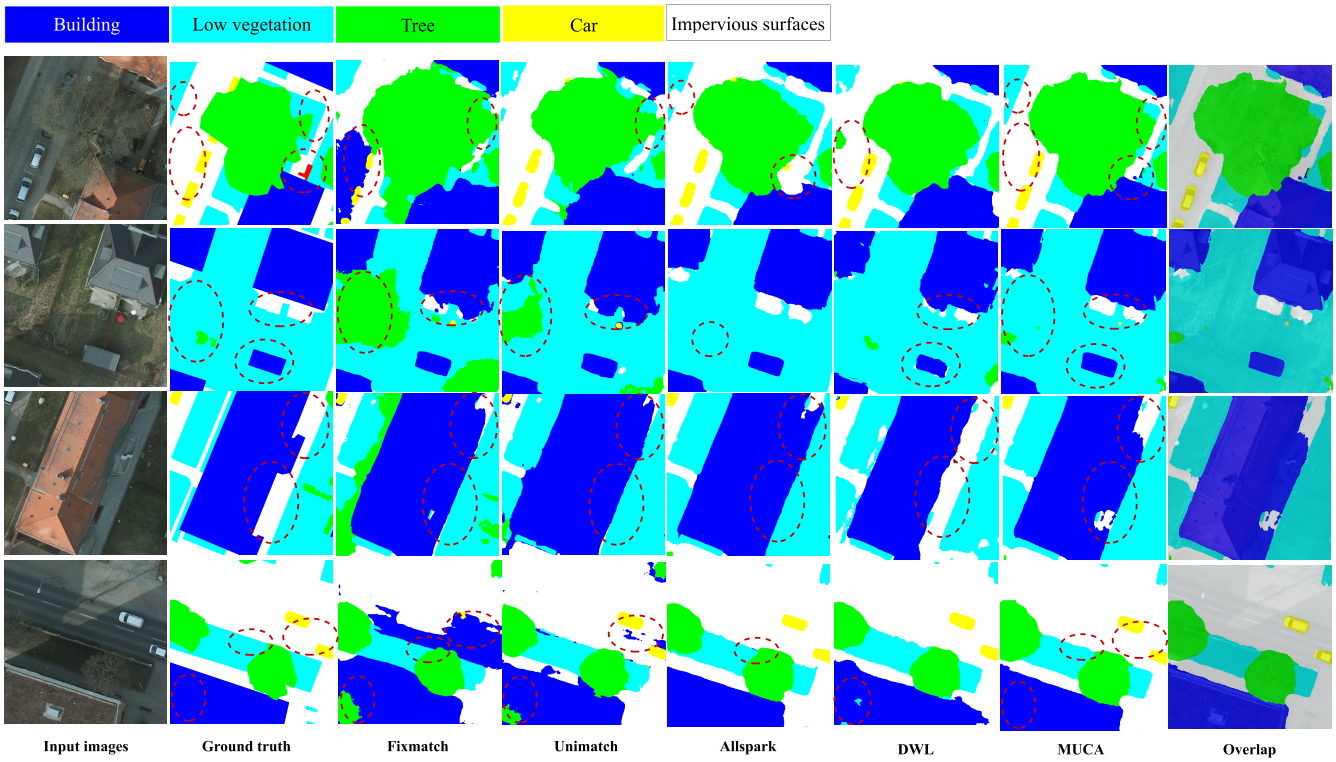


Fig. 7. Visual comparison of semantic segmentation results with different semisupervised methods on the ISPRS-Potsdam dataset.

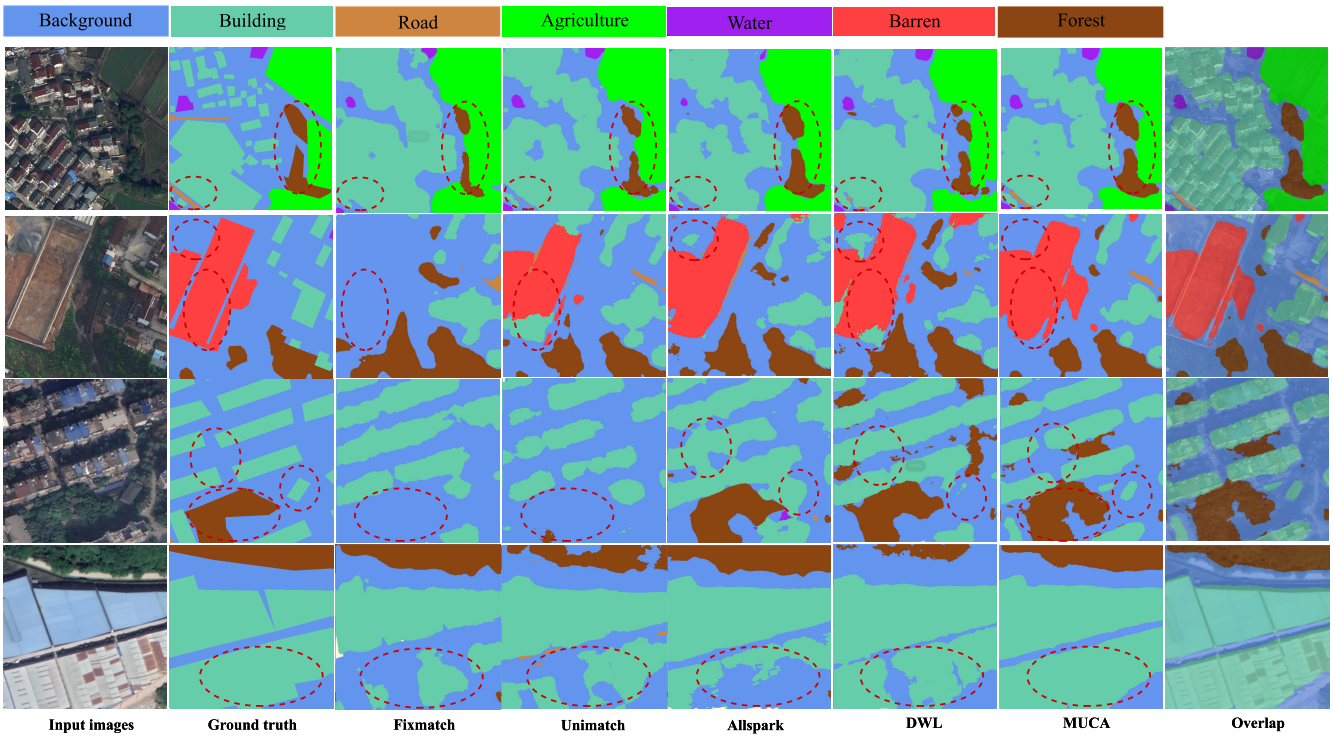


Fig. 8. Visual comparison of semantic segmentation results with different semisupervised methods on the LoveDA dataset.

Building, Barren, and Background classes. While Allspark and DWL models achieve relatively precise segmentation boundaries, they still demonstrate poor classification performance in mixed zones containing both Building and Barren classes. Furthermore, the segmentation accuracy for individual structures within the Building class remains suboptimal, failing to delineate architectural details effectively. In contrast, our proposed MUCA shows significant advantages in accurately

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TABLE VII
MODEL GENERALIZABILITY EXPERIMENTS

Dataset	Network	Model	mIoU(5%)
ISPRS-Potsdam	U-Net	Onlysup	57.52
		NoUC	61.59
		MUCA	64.21
	PSPNet	Onlysup	63.46
		NoUC	66.16
		MUCA	68.87
	DeepLabv3+	Onlysup	66.83
		NoUC	68.01
		MUCA	70.49
	SegFormer-B2	Onlysup	72.01
		NoUC	73.14
		MUCA	74.62
LoveDA	U-Net	Onlysup	38.12
		NoUC	40.63
		MUCA	42.32
	PSPNet	Onlysup	40.16
		NoUC	42.98
		MUCA	44.56
	DeepLabv3+	Onlysup	41.45
		NoUC	43.87
		MUCA	45.98
	SegFormer-B2	Onlysup	44.20
		NoUC	48.21
		MUCA	50.97

identifying Building, Agriculture, Barren, and Background classes. Finally, we overlay the segmentation results with the original images in the last column to clearly demonstrate our advantages. This overlayed visualization indicates that our model achieves superior visual results on the LoveDA dataset compared with other models.

V. DISCUSSION AND CONCLUSION

Our study addresses the challenges associated with semi-supervised RS image semantic segmentation by proposing the MUCA model. The MUCA model includes two special modules, i.e., MSUC and CTSA. The goal of MSUC is to learn rich multiscale information; meanwhile, CTSA distinguishes the high interclass similarities through the cross-network attention mechanism. The new features constructed by CTSA enable the student network decoder to benefit from the dual enhancement of both WA and SA, achieving better optimization results and demonstrating greater stability during the training phase.

Compared with SOTA algorithms, our method achieved the best results for the metrics mIoU, mF1, and Kappa; however, it did not achieve the best IoU for all the categories. This phenomenon is understandable and explainable. On the one hand, this article does not address other problems in the semi-supervised domain, such as long-tailed distribution among classes. On the other hand, different models have different focuses, which may lead to a particularly good result for one class and a poor result for others. This study highlights a more realistic scenario where MUCA demonstrates the best overall performance across all classes. However, current semi-supervised models for RS image analysis still face several bottlenecks. For instance, segmentation performance remains suboptimal in areas with mixed land cover, and visualized results on the LoveDA dataset exhibit flawed boundaries for classes such as Forest and Road. Notably, neither our proposed MUCA nor existing semi-supervised models achieve an mIoU exceeding 52% on LoveDA, indicating substantial room for improvement. Furthermore, research on semi-supervised models for complex spectral imaging (e.g., multispectral) and multimodal RS data fusion remains limited. Our future work will focus on cross-modal alignment and developing semi-supervised models that integrate multisource data, including multispectral imagery, SAR, and LiDAR.

Finally, we hope that the model proposed in this article can serve as a simple and powerful baseline in the field of semi-supervised semantic segmentation of RS images. We also aim to inspire more valuable research for future work.

ACKNOWLEDGMENT

The authors would like to express their sincere appreciation to the anonymous reviewers for their insightful comments, which have greatly aided them in improving the quality of this article.

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